

# Vector Spaces or Linear Spaces

In our quest to generalize we discovered many types of things that behaved the same way. Their structure was the same even though the context was vastly different: Homogeneous linear ODEs, homogeneous linear algebraic systems, linear operators, etc...

All these and many many more things share the same structure. Once we write them down and study the similarities, we can then apply the machinery we learned here to all the other things that share that same structure! The power of abstraction is limitless! (No pun intended)

A vector space (or linear space)  $V$  is a nonempty collection of objects called vectors defined over a field  $\mathbb{R}$  (or  $\mathbb{C}$ ) if:

Closure Properties:

- ① Given any two  $\vec{x}, \vec{y} \in V$  there is a rule called  $+$  (addition) that gives a unique element  $(\vec{x} + \vec{y}) \in V$  called the sum of  $\vec{x}$  and  $\vec{y}$ .
- ② Given any element  $\vec{x} \in V$  and a scalar  $\lambda \in \mathbb{R}$  (or  $\mathbb{C}$ ) there is a rule called scalar multiplication that gives a unique element  $\lambda \vec{x} \in V$ .

Addition has the following rules:

- ③  $\vec{x} + \vec{y} = \vec{y} + \vec{x}$ , for all  $\vec{x}, \vec{y} \in V$  Commutivity
- ④  $(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$  for all  $\vec{x}, \vec{y}, \vec{z} \in V$  Associativity
- ⑤ There exists  $\vec{0} \in V$  s.t.  $\vec{x} + \vec{0} = \vec{x}$  zero vector
- ⑥ For all  $\vec{x}$  there exists  $\vec{y} \in V$  s.t.  $\vec{x} + \vec{y} = \vec{0}$  Inverse

Scalar multiplication has the following rules:

- ⑦  $1 \cdot \vec{x} = \vec{x}$  for all  $\vec{x} \in V$
- ⑧  $\alpha(\beta \vec{x}) = (\alpha\beta) \vec{x}$  for all  $\vec{x} \in V$  and  $\alpha, \beta \in \mathbb{R}$  (or  $\mathbb{C}$ )
- ⑨  $(\alpha + \beta) \vec{x} = \alpha \vec{x} + \beta \vec{x}$  for all  $\vec{x} \in V$  and  $\alpha, \beta \in \mathbb{R}$  (or  $\mathbb{C}$ )
- ⑩  $\alpha(\vec{x} + \vec{y}) = \alpha \vec{x} + \alpha \vec{y}$  for all  $\vec{x}, \vec{y} \in V$  and  $\alpha \in \mathbb{R}$  (or  $\mathbb{C}$ )

Note: Rules 1, 4, 5, and 6 form what is called a "group" w/ operation  $+$ . Rule 3 makes this "group" commutative, or what is called "Abelian."

Note: "Vector" here can mean anything that satisfies the above 10 properties, be they arrows, matrices, numbers, sandwiches, planes, lines, ...

Theorem: The zero vector is unique.

Pf: HW, it's not difficult!

Theorem: Every element in a vector space has a unique (additive) inverse.

Pf: Suppose  $\vec{x} + \vec{y}_1 = \vec{0}$  and  $\vec{x} + \vec{y}_2 = \vec{0}$ . Then

$$\vec{y}_2 = \vec{y}_2 + \vec{0} = \vec{y}_2 + (\vec{x} + \vec{y}_1) = (\vec{y}_2 + \vec{x}) + \vec{y}_1 = \vec{0} + \vec{y}_1 = \vec{y}_1 \quad \checkmark$$

Theorem: The relation  $0\vec{x} = \vec{0}$  is always true in a vector space.

Pf: Consider  $0\vec{x} + 1\vec{x} = (0+1)\vec{x} = 1\vec{x}$ .

Since every element has an additive inverse, say  $\vec{x} + \vec{y} = \vec{0}$

$$\begin{aligned} 0\vec{x} + 1\vec{x} + \vec{y} &= 1\vec{x} + \vec{y} = \vec{0} \\ \Rightarrow 0\vec{x} &= \vec{0} \quad \checkmark \end{aligned}$$

Theorem:  $\vec{y} = (-1)\vec{x}$  acts like the additive inverse to  $\vec{x}$ .

Pf: Suppose  $\vec{x} + \vec{y} = \vec{0}$ , then  $(-1)\vec{x} = (-1)\vec{x} + \vec{x} + \vec{y} = (-1+1)\vec{x} + \vec{y} = 0\vec{x} + \vec{y} = \vec{0} + \vec{y} = \vec{y} \quad \checkmark$   
 $\therefore (-1)\vec{x} = \vec{y} \quad \checkmark$

We can combine conditions ① & ② together for simplicity and get:

Closure property ①:  $\alpha \vec{x} + \beta \vec{y} \in V$  whenever  $\vec{x}, \vec{y} \in V$  and  $\alpha, \beta \in \mathbb{R}$  (or  $\mathbb{C}$ ).

## Examples of Vector Spaces

Ex//  $\mathbb{R}$  is a vector space over itself. (Real numbers as vectors)

Ex//  $\mathbb{C}$  is a vector space over itself. (Complex numbers as vectors)

Ex// Any field is a vector space over itself. (Scalars as vectors)

Ex//  $\mathbb{R}^n$  is a vector space over  $\mathbb{R}$ . ( $\vec{x} = [x_1, x_2, \dots, x_n]$  or  $\vec{x}^T$  as vectors)

( $n \times 1$  or  $1 \times n$  vectors as vectors)

Ex//  $\mathbb{C}^n$  is a vector space over  $\mathbb{C}$  ( $n \times 1$  or  $1 \times n$  vectors as vectors).

Ex//  $M_{mn}$ : the space of all  $m \times n$  matrices over  $\mathbb{C}$ .

confirm property 1, 2, and a random one of classes choosing

Ex// The space of continuous functions on  $[0, 1]$  over  $\mathbb{R}$ .

Here,  $f$  &  $g$  are vectors.

$$① (f+g)(t) = f(t) + g(t) \quad \forall t \in [0, 1]$$

$$② (cf)(t) = c f(t)$$

etc..

Ex// Solution spaces of linear homogeneous  $n$ th order ODEs

They satisfy the superposition principle (proven), so by default if  $y_1$  &  $y_2$  solve an ODE,

so does  $\alpha y_1 + \beta y_2$ .

Check all other properties.

Ex// Solution spaces of nonhomogeneous ODEs is not a vector space.

Why not? Should be trivial now, make sure you know why if it's not trivial!

Ex//  $P_3$ : all polynomials of degree 3 or lesser on  $\mathbb{C}$  over  $\mathbb{R}$ .

I.e.  $a_i, b_i \in \mathbb{C}, t \in \mathbb{R}$

$$p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 \quad t \in \mathbb{R}.$$

$$q(t) = b_0 + b_1 t + b_2 t^2 + b_3 t^3$$

## Vector Subspaces

Sometimes subsets of vector spaces are important vector spaces in their own right!

Defn: A nonempty subset  $W$  of a vector space  $V$  is a subspace of  $V$  if it is closed under addition and scalar multiplication:

$$(i) \text{ if } \vec{u}, \vec{v} \in W, \text{ then } \vec{u} + \vec{v} \in W$$

$$(ii) \text{ if } \vec{u} \in W \text{ and } \lambda \in \mathbb{R} \text{ (or } \mathbb{C}), \text{ then } \lambda \vec{u} \in W$$

We don't need to check the other conditions, the ambient vector space  $V$  guarantees they'll be satisfied.

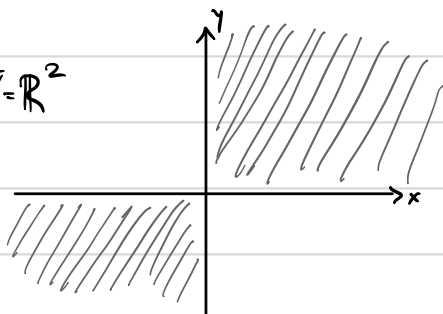
Can also combine the above into one condition:  $\alpha \vec{u} + \beta \vec{v} \in W$  for all  $\vec{u}, \vec{v} \in W$  and  $\alpha, \beta \in \mathbb{R}$  (or  $\mathbb{C}$ ).

Note: It is absolutely important that  $\vec{0} \in W$ !

Since  $\vec{u} \in W$  and  $\lambda \vec{u} \in W$  for any  $\lambda \in \mathbb{R}$  (or  $\mathbb{C}$ ), then

$$\vec{0} = \vec{u} + (-1)\vec{u} \in W$$

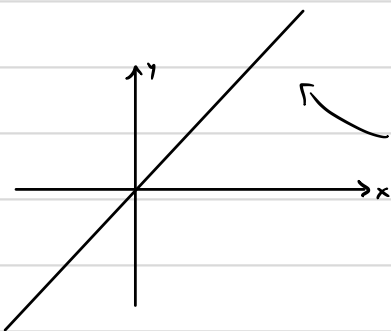
Ex //  $V = \mathbb{R}^2$



$$W = \{(x, y) \mid \text{s.t. } xy \geq 0\} \quad \leftarrow \text{First \& Third quadrant.}$$

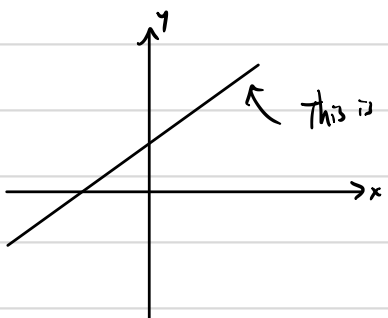
This is not a vector subspace. Why not?

Ex //  $V = \mathbb{R}^2$



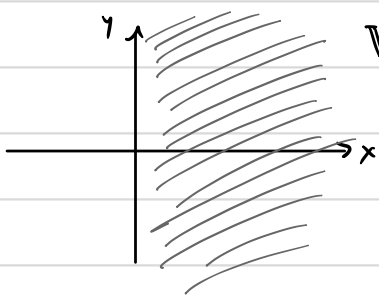
This is a vector subspace of  $\mathbb{R}^2$ . Why?

Ex //  $V = \mathbb{R}^2$



This is not a vector subspace of  $\mathbb{R}^2$ . Why not?

Ex //  $V = \mathbb{R}^2$



$W = \{(x, y) \mid x \geq 0\}$  Is not a vector subspace of  $\mathbb{R}^2$ . Why not?

Ex // The solutions to  $A\underline{x} = \underline{0}$  is a very important subspace called the Null Space (or Kernel).

$$\text{If } A\underline{x} = \underline{0} \text{ and } A\underline{y} = \underline{0}, \text{ then } A(\alpha\underline{x} + \beta\underline{y}) = \alpha A\underline{x} + \beta A\underline{y} = \alpha \underline{0} + \beta \underline{0} = \underline{0} \quad \checkmark$$

This space will have an extremely important role to play very soon. The "dimension" of this space is called the Nullity.  
It's part of the very important "Rank-Nullity Theorem."

*not defined yet!*

Ex// The collection of all  $n \times n$  matrices forms a vector space. The collection of matrices w/ zero trace forms a subspace.

pf:  $\text{Tr}(\alpha A + \beta B) = \sum_{i=1}^n \alpha a_{ii} + \beta b_{ii} = \alpha \sum_{i=1}^n a_{ii} + \beta \sum_{i=1}^n b_{ii} = \alpha(0) + \beta(0) = 0 \quad \checkmark$

Definition: (orthogonal complement) Let  $V$  be a subspace of  $\mathbb{R}^n$ . A vector  $\vec{u}$  is orthogonal to subspace  $V$  provided that  $\vec{u}$  is orthogonal to every vector in  $V$ .

The set of vectors orthogonal to  $V$  is called the orthogonal complement of  $V$

$$V^\perp = \{ \vec{u} \in \mathbb{R}^n \mid \vec{u} \cdot \vec{v} = 0 \text{ for every } \vec{v} \in V \}$$

Ex// The orthogonal complement of a vector space is a vector subspace of  $\mathbb{R}^n$ .

pf: HW for you.

Def'n: A linear combination of vectors  $\{v_1, v_2, \dots, v_n\}$  is an expression of the form:

$$c_1 v_1 + c_2 v_2 + \dots + c_n v_n$$

If all  $v_i$ 's are column vectors this may be rewritten as

$$A \vec{c} = \begin{pmatrix} | & | & \dots & | \\ v_1 & v_2 & \dots & v_n \\ | & | & \dots & | \end{pmatrix} \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = c_1 \begin{pmatrix} | \\ v_1 \\ | \end{pmatrix} + c_2 \begin{pmatrix} | \\ v_2 \\ | \end{pmatrix} + \dots + c_n \begin{pmatrix} | \\ v_n \\ | \end{pmatrix}$$

A column picture.

If all  $c_i$ 's were zero, the linear combination would obviously be zero. The main question here is: Are there any other numbers that would make the linear combination vanish?

Definition: If only the trivial combination gives zero, so that  $c_1 v_1 + \dots + c_n v_n = 0$  only happens when  $c_1 = c_2 = \dots = c_n = 0$  then the vectors  $v_1, \dots, v_n$  are linearly independent.

Otherwise they are linearly dependent, and one of them is a linear combination of the others.

Note: The vectors  $v_i$  don't need to be arrows, or row/column vectors! They may be continuous functions,  $3 \times 4$  matrices, etc...

Ex// If one of the vectors is the zero vector, the whole collection is automatically dependent.

over  
→

Sub ex/  $c_1 \vec{0} + c_2 \vec{v}_2 + \dots + c_n \vec{v}_n = \vec{0}$

when  $c_1 \in \mathbb{R}$  and  $c_2 = c_3 = \dots = c_n = 0$

So linear dependence guaranteed.

Ex/ The columns of the matrix  $\begin{bmatrix} 1 & -1 & 3 & 3 \\ 2 & -7 & 6 & 1 \\ 4 & 2 & -6 & 0 \end{bmatrix}$

are dependent since the last column is the sum of the first three. Thus,

$$(1) \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} + (1) \begin{pmatrix} -1 \\ -7 \\ 2 \end{pmatrix} + (1) \begin{pmatrix} 3 \\ 6 \\ -6 \end{pmatrix} + (-1) \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Since there is a way to get a linear combination to sum to zero, the collection is dependent.

Ex/  $\begin{pmatrix} 3 & 9 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 7 \end{pmatrix}$  has linearly independent columns. Note that

$$c_1 \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 9 \\ 2 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 4 \\ 6 \\ 7 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{exactly when}$$

$$c_3 = 0 \quad \text{since} \quad 7c_3 = 0$$

$$c_2 = 0 \quad \text{since} \quad 2c_2 + 6c_3 = 0, \text{ and}$$

$$c_1 = 0 \quad \text{since} \quad 3c_1 + 9c_2 + 4c_3 = 0$$

For column vectors, then, the null space (or kernel) must be  $\{\vec{0}\}$ , which is equivalent to saying the columns are independent, and that the system

$$\begin{pmatrix} 3 & 9 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 7 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

contains only the trivial solution, which means  $A\vec{c} = \vec{0}$ , etc...

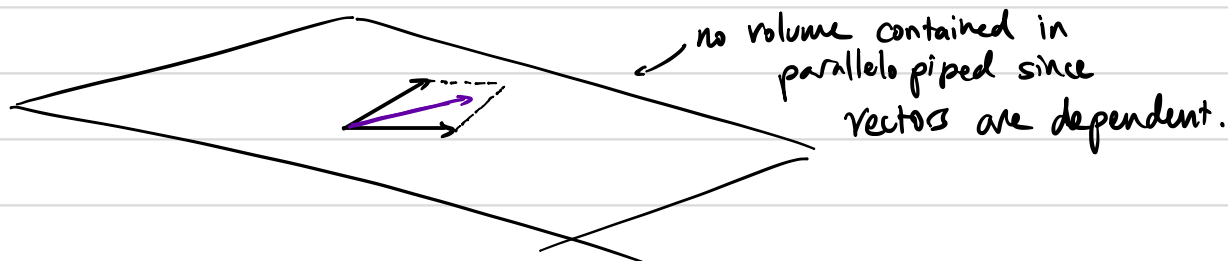
Visually, what is going on?

If given an  $n \times n$  system, consider the equation  $A\vec{c} = \vec{0}$ .

This system has a unique solution of  $\vec{c} = \vec{0}$  iff  $A$  is invertible. By the I.M.T. this means the matrix  $A$  is row/column equivalent to the identity matrix.

Moreover,  $\det(A) \neq 0$ . That means each vector in the columns of  $A$  "point" in a different direction since the volume they span is not zero!

If the collection of vectors were linearly dependent then one column would be a linear combination of the others. This means that column would not end up a pivot column after putting the matrix in RREF. That also means that vector would be pointing in the same direction as other vectors, so  $\det(A) = 0$



Ex// Consider the collection of vectors in  $[0, 2\pi]$   $\{1, \cos(2x), \sin^2(x)\}$ .  
This collection is dependent since

$$\left(\frac{1}{2}\right) 1 + \left(-\frac{1}{2}\right) \cos(2x) + (-1) \sin^2(x) = 0$$

due to the double-angle identity in trigonometry.

Ex/ Suppose we had any two differentiable functions  $\phi_1$  &  $\phi_2$ . We wish to tell if they are dependent. Thus, we need  $c_1$  and  $c_2$  s.t.

$$c_1 \phi_1 + c_2 \phi_2 = 0.$$

But this alone is not enough to determine  $c_1$  and  $c_2$ . We need another condition involving  $c_1$  &  $c_2$ . If we differentiate both sides of the equation, this gives us the condition we need!

$$c_1 \phi_1 + c_2 \phi_2 = 0$$

$$c_1 \phi_1' + c_2 \phi_2' = 0$$

$$\text{or } \begin{pmatrix} \phi_1 & \phi_2 \\ \phi_1' & \phi_2' \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This system has a unique solution of  $\vec{c} = \vec{0}$  iff  $\phi_1 \phi_2' - \phi_2 \phi_1' \neq 0$   
(from Cramer's Rule).